高分子化学学习笔记

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1 Introduction

Polymer chemistry is the foundation of polymer science, studying the molecular design, synthesis, and modification of polymer compounds, and researching polymerization reactions and the principles of polymer chemical reactions. It is responsible for providing new chemical compounds for polymer science research and supplying new materials and synthesis methods for the national economy.

Polymer Physics is the theoretical foundation of polymer science. It mainly studies the aggregated state structure, properties, and characterization of polymeric machines, as well as the relationships between structure and properties, and the influence of structure on external fields. It guides the molecular design of polymer compounds and the rational use of polymers as materials, serving as a bridge between synthesis and application.

Polymer engineering studies and designs polymer reaction engineering, polymer molding processes, and the corresponding theoretical and methodological research, serving as the connection between polymer science and the polymer industry.

$$\begin{tikzcd}
\text{合成} \arrow[r, "\text{（高分子合成）}"] & \text{结构} \arrow[d] \\
\text{应用} \arrow[u, dashed] & \text{性能} \arrow[l, "\text{（高分子成型）}"]
\end{tikzcd}$$

高分子化学是研究聚合物合成与化学反应的一门学科，涉及到两类反应：
聚合物的合成反应

$$\text{小分子} \rightarrow \text{大分子}$$

聚合物的化学反应

$$\text{新的大分子} \leftarrow \text{大分子} \rightarrow \text{小分子}$$

- 反应的机理
- Factors affecting the reaction
- 反应的动力学
- Implementation method of the reaction

1.1 The Basic Concepts of Polymers

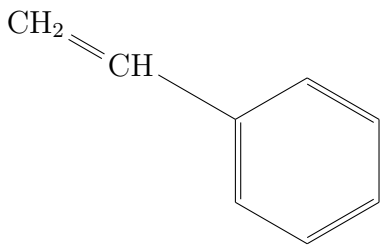
1.1.1 Definition of Polymers

Polymers, also known as macromolecules, are large molecular compounds with a very high relative molecular mass, formed by many identical simple structural units connected repeatedly by covalent bonds. There is no clear boundary for how large a molecule must be to be considered a macromolecule.

一般的高分子分子量在 10^6—10^7 之间。

- Low molecular weight raw materials for synthesizing polymers from monomers.

  Nylon-66 monomer



苯乙烯

CH_2 ~~CH~~ Cl 氯乙烯

- 重复结构单元

  The smallest basic unit on a macromolecular chain, whose chemical groups and structure can repeatedly

- 结构单元

  Monomer molecules become the basic units of macromolecular chains after polymerization. The elemental composition of the structural units can be the same as that of the monomers, or it can be different.

- 单体单元是指分子通过聚合反应形成的，其元素组成与单体完全相同的结构单元

- 聚合度

  A measure of the size of polymer molecules

1.2 Classification and Nomenclature of Polymers

1.2.1 Classification of Polymers

- Classified according to the performance and use of the materials

  Rubber, plastic, fiber, coating, adhesive, functional polymer

- Classification based on the main chain structure of polymers

  Carbon chain polymers, copolymers, elemental organic polymers

- Classification based on the morpholo gical changes of polymers after heati ng Thermoplastic polymers Thermos etting polymers

1.2.2 Naming of Polymers

- Classified according to the performance and use of the materials
- Classification based on the main chain structure of polymers
- Classification according to the morphological changes of polymers after heating

1.2.3 聚合反应

Definition: The process of obtaining polymers from small molecule monomers through chemical method

（1）Classified according to the composition and structural changes of monomers and polymers

- Addition reaction
- Condensation reaction
- The difference between addition polymerization and condensation polymerization

（2）按聚合机理分类

- 连锁聚合
- 逐步聚合
- Comparison of Chain Polymerization and Step-Growth Polymerization

1.3 Basic Characteristics of Macromolecular Compounds

1.3.1 The main chain of macromolecules is mostly connected by rep eating structural units in the form of covalent bonds 1.3.2 Large mol ecular weight 1.3.3 Molecular weight is polydisperse 1.3.4 Molecul ar morphology is diverse 1.3.5 Has significant multi-level structure

1.4 聚合热

Thermodynamics discussion scope: the feasibility of reactions; the direction of reactions; reaction equilib

1.4.1 General Concepts of Thermodynamics

聚合热（焓）因单体转变为聚合物是体系的无序度减小故 $\Delta S < 0$，而且各单体的聚合熵一般变化不大约为 $-105 \sim -125 jmol^{-1}K^{-1}$ 因此要使聚合反应进行必须

$$\Delta H < 零$$

$$|\Delta H| > |T\Delta S|$$

- Polymerization reactions must be exothermic reactions
- The amount of heat released varies greatly. The greater the polymerization heat, the greater the tendenc

The polymerization heat of olefin monomers is such that what is broken is a carbon-carbon double bond, and what is reorganized are two carbon-carbon single bonds, so the polymerization heat is $-\Delta H = 90\ kjmol^{-1}$. The structure of the monomer also affects the polymerization heat, usually

- The larger the bond energy of the monomer, the smaller the polymerization heat.
- The greater the bond energy of the polymer, the greater the polymerization heat.

1) Steric effect of substituents

1,1-二取代的单体聚合热较小，聚合倾向较小

2) Conjugation effect and hyperconjugation effect

Conjugation effects and hyperconjugation effects make the bond energy of the monomer larger, the po

3) 取代基的电负性

The greater the electronegativity of the substituent, the smaller the bond energy of the monomer, the greater the polymerization heat, and the greater the polymerization tendency. 4) Hydrogen bonding and solvation effects

Hydrogen bonding and solvation cause the monomer to have higher bond energy, lower polymerizatio

Polymerization heat of condensation monomers

1) The total internal energy does not change much before and after polymerization, and the actual polymeriza

2) The monomers of condensation reactions have high polarity, and the molecular aggregation is relatively regular. 3) Polymers can only be formed when the number of functional groups in each participating molecule is greater than or equal to 2; this is a condensation reaction. 4) Condensation reaction is the process of forming polymers through multiple condensation reactions between functional groups.

Heat of polymerization of cyclic monomers In ring-opening polymerization, the types of bonds do not change before and after polymerization, so the heat of polymerization mainly comes from the release of ring strain.

Circumferential tension mainly includes two categories:

- Torsional stress caused by key angle deformation
- Conformational strain caused by repulsion between hydrogen or substituents

For cycloalkanes, the ring-opening polymerization ability of different ring systems can be summarized as

1) 三、四元环

The angle strain and polymerization heat are both large, making ring-opening polymerization easy to c

2) Five-membered ring

The bond angle is about 108°, the angular tension and ΔH are both small, and the ΔS term plays an in

3) Six-membered ring

Usually presents chair or boat conformations, with bond angle deformation approaching zero, $\Delta H \approx 0$, ΔG positive, therefore generally unable to polymerize.

4) Eight-membered or larger ring

The hydrogen or substituents within the ring are in a crowded state, and due to repulsion, transannular

5) Rings over eleven yuan

Cross-ring tension is basically gone, and the contributions of ΔH and ΔS are similar.

The ease of ring opening of cycloalkanes can be expressed as:

$$3, 4 > 8 > 7, 5$$

2 Condensation and Step-Growth Polymerization

2.1 Introduction

Condensation reaction refers to a reaction between organic functional groups in which, in addition to forming the main product, a small molecule is generated.

- In the reactants participating in the condensation reaction, each molecule has only one functional group
- Condensation reactions can only form low molecular weight compounds
- A polymer can only be formed in a condensation reaction when the number of functiona l groups in each reacting molecule is greater than or equal to 2.
- The polycondensation reaction is the process of forming polymers through multiple condensation reacti

2.1.1 基于步骤的聚合基本概念

Step-growth polymerization is one of the most fundamental types of polymer synthesis. T he basic characteristic of step-growth polymerization is the reaction between functional groups . The degree of polymerization gradually increases over time, and a high conversion rate can b e achieved when monomers are depleted, thus exhibiting different patterns from chain polymer ization.

2.1.2 Stepwise Polymerization Types

（1) Condensation reaction: A condensation polymerization reaction between functional groups, with the

（2）聚加成反应：形式上是加成反应，机理是逐步的。

（3) Open-loop response: part is a step response

（4）Oxidative coupling: the condensation reaction of monomers with oxygen.

（5）Diels-Alder 反应

Table 1: Comparison of Step-Growth and Chain-Growth Polymerization

连锁聚合（Chain Polymerization）	逐步聚合（Stepwise Polymerization）
需活性中心：自由基、阳离子或阴离子，有链引发、增长、转移、终止等基元反应.	官能团间反应，无特定的活性中心，无链引发、增长、终止等基元反应.
单体一经引发，便迅速连锁增长，各步反应速率和活化能差别很大.	各反应逐步进行，每一步反应速率和活化能大致相同.
体系中只有单体和聚合物，无分子量递增的中间产物.	体系含单体和一系列分子量递增的中间产物.
转化率随着反应时间而增加，分子量变化不大.	分子量随着反应的进行缓慢增加，而转化率在短期内很高.

2.2 Condensation Reaction

官能团之间经过多次缩合形成聚合物的反应

- Polymers have characteristic structural functional groups
- There is the formation of low molecular by-products
- The molecular weight of the condensate and the monomer is not an integer multiple

2.2.1 Degree of Functionality and Average Degree of Functionality

Functional group: A group of atoms in a monomer molecule that can participate in reactio ns and exhibit types of reactions. — Functionality: The number of functional groups that partic ipate in reactions.

Average functionality When two or more types of monomers are mixed for a polyconden sation reaction, the ratio of the number of functional groups in the reaction system that can par ticipate in the reaction to the total amount of monomers. It has a great impact on the polyconde nsation product.

$$f = 1, \text{产物为低分子}$$

$$f = 2, \text{products are linear polymers}$$

$$f > 2, \text{products are branched or bulky polymers}$$

2.2.2 Classification of Condensation Reactions

a) Classification according to the molecular structure of the product

- Linear condensation reaction
- Polycondensation reaction of body size

b) Classified according to the types of monomers participating in the reaction

- Self-condensation
- co-polycondensation

c) Classify according to the thermodynamic characteristics of the reaction

- Balance condensation
- Unequal condensation

2.3 Linear Condensation Reaction Mechanism

2.3.1 Linear Condensation and Cyclization Reactions

In the polycondensation reaction process, there are often concurrent cyclization reactions, including intra

Cyclization reactions occur within the same monomer unit, and intramolecular cyclization occurs between different monomer units within the same molecule. Cyclization reactions reduce the degree of polymerization and affect the properties of the product.

2.3.2 Linear Condensation Mechanism

One of the important characteristics of condensation reactions is that the growth of macrom olecular chains occurs step by step. For example, using aAa to represent a diol and bBb, the proc ess by which they form polyester is as follows:

Two types of molecular monomers first react with each other to form a dimer

$$aAa + bBb \rightleftharpoons a\left[A-B\right]b + ab$$

The dimer continues to react with the monomer to form a trimer

$$a\left[A-B\right]b + bBb \rightleftharpoons a\left[A-B\right]b\left[A-B\right]b + ab$$

Or dimers react with each other to form tetramers

$$a\left[A-B\right]b + a\left[A-B\right]b \rightleftharpoons a\left[A-B\right]b\left[A-B\right]b + ab$$

By analogy, polymer chains are gradually formed. This process can be represented as

$$n\text{聚体} + m\text{聚体} \rightleftharpoons (n+m)\text{聚体} + ab$$

It is meaningless to study the conversion rate of the polycondensation reaction. We use the reaction extent P t

$$P = \frac{n_0 - n}{n_0}$$

其中 $n_0 - n$ 表示反应过程中消耗的基团数，n_0 表示反应开始时基团的总数。聚合度与反应程度的关系为

$$\bar{X}_n = \frac{1}{1-P}$$

Another characteristic of condensation reactions is the reversibility of the reaction equilibrium.

- k small, polyester, small molecules have a greater impact on the degree of polymerization.
- In k, polyamides, small molecules have some impact on the degree of polymerization.
- k large, can be considered irreversible, phenolic resin system.

When performing a condensation reaction in a closed system, the concentration of small molecules in the reaction system gradually increases, causing the reaction to reach an equilibrium state and limiting further growth of the polymerization degree. Therefore, in practical operation, the reaction is often driven toward the formation of high polymers by continuously removing the generated small molecules (such as water, alcohol, etc.), thereby achieving a higher polymerization degree.

For example, in polyester production, the yield depends on the vacuum level.

Step-growth characteristics are common to condensation reactions, and the degree of reversibility depend

- Polymer chains grow step by step
- The reaction shows a reversible equilibrium
- There are often side effects

2.3.3 Side Reactions in Condensation Process

1) Group elimination reaction

Dicarboxylic acids will decarboxylate when heated, causing changes in the number of functional groups.

2) Chemical degradation

Condensation reactions are reversible. Monomers often act as degradants of polymers, resulting in a decrease

3) Chain exchange reaction

The new functional groups formed in the condensation reaction are highly reactive and can easily react with functional groups in other molecules, leading to chain exchange reactions. Chain exchange of the same condensation product results in a narrower molecular weight distribution, while chain exchange of different condensation products results in block copolymers.

2.4 Linear Condensation Reaction Kinetics

2.4.1 Functional Groups and Other Reactivity Concepts

In the process of forming macromolecules, polycondensation reactions proceed step by step. If each step has a different reaction rate constant, the kinetic analysis of polycondensation reactions will be very complicated. To simplify the analysis, the concept of functional group equivalence activity was proposed, that is, it is assumed that all functional groups involved in the reaction have the same reactivity. This assumption makes the kinetic analysis of polycondensation reactions feasible and can describe the reaction process well in many practical cases.

Flory 提出了官能团等活性假设，认为官能团之间的碰撞次数和有效碰撞几率与高分子的扩散速率无关，体系粘度增大时虽然整个高分子运动速率减慢，但链段运动和链段的官能团活动并未受到限制。由于高分子的活动迟缓，扩散速率低反而是官能团之间碰撞时间变长，有利于提升反应速率。

Functional group and other activity hypotheses The theory of functional group activity is approximate, n

2.4.2 Irreversible Condensation Kinetics

If the low molecular by-products in the system are continuously removed, the reaction proceeds forward. It can be regarded as an irreversible reaction. Take polyester reaction as an example.

$$-\frac{d[COOH]}{dt} = \frac{k_1 k_3 [COOH][OH][H^+]}{k_2 k_{HA}}$$

Exogenous acid catalytic system

The esterification rate of the self-catalyzed polycondensation reaction is too low. Acid catalysts are often used. During the polymerization process, the concentration of hydrogen ions can be regarded as constant, formula

$$-\frac{d[COOH]}{dt} = \frac{k_1 k_3 [COOH][OH][H^+]}{k_2 k_{HA}}$$

All the k's here can be combined with [H^+] into a single constant k', resulting in

$$-\frac{dc}{dt} = k'c^2$$

变化可得

$$\frac{1}{c} - \frac{1}{c_0} = k't$$

Introducing the reaction extent p, we get

$$\frac{1}{1-p} - 1 = k'c_0 t$$

The above formula shows that the degree of polymerization of the acid-catalyzed polycondensation reaction changes rapidly with polymerization time. Generally, the rate constant of acid-catalyzed polycondensation reaction is 1-2 orders of magnitude larger than that of the self-catalyzed polycondensation reaction.

自催化体系

When the number of functional groups in a dibasic acid and a diol are the same, whether or not externally added acidic hydrogen ions are present, they all come from the dibasic acid itself, and two cases are discussed:

1) Carboxylic acids do not ionize

At this time, the polyesterification reaction forms a bimolecular complex from two molecules of carboxylic acid, while participating in condensation with one molecule of hydroxyl to become a tertiary reaction. The rate equation is

$$-\frac{dc}{dt} = k''c^3$$

Points earned

$$\frac{1}{2}\left(\frac{1}{c^2} - \frac{1}{c_0^2}\right) = k''t$$

Introducing the reaction extent p, we get

$$\frac{1}{(1-p)^2} - 1 = 2k''c_0^2 t$$

2.4.3 Balance Condensation Kinetics

The equilibrium constant for the polyesterification reaction is very small. If the small molecule by-products are not removed in time, the reverse reaction cannot be ignored. Let the initial concentration of the carboxyl group be c_0, and the concentration at time t be c, considering two cases: water is not removed and water is completely removed.

$$-\mathrm{COOH} + -\mathrm{OH} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} -\mathrm{OCO}- + \mathrm{H_2O}$$

起始	c_0	c_0	0	0
t 时（水不排除）	c	c	$c_0 - c$	$c_0 - c$
t 时（水部分排除）	c	c	$c_0 - c$	n_w

Closed system, water is not eliminated

$$-\frac{dc}{dt} = -k_1 c^2 + k_{-1}(c_0 - c)^2$$

Plug in c=$c_0(1-p)$, we get

$$\frac{dp}{dt} = k_1 c_0[(1-p)^2 - \frac{p^2}{K}]$$

When K is large, the second term can be ignored. Then the integral is

$$\frac{1}{1-p} - 1 = k' c_0 t$$

Partial water exclusion is

$$\frac{dc}{dt} = -k_1 c^2 + k_{-1}(c_0 - c) n_w$$

Substitute c=$c_0(1-p)$, we get

$$\frac{dp}{dt} = k_1[c_0(1-p)^2 - \frac{p n_w}{K}]$$

Similarly, when K is large, the second term can be ignored. Then the integral is

$$\frac{1}{1-p} - 1 = k' c_0 t$$

2.5 线性缩聚物的聚合度

2.5.1 Effect of Reaction Extent and Equilibrium Constant on Degree of Polymerization

When the system is closed, at equilibrium, let

$$\frac{dp}{dt} = k_1 c_0[(1-p)^2 - \frac{p^2}{K}] = 0$$

则

$$p = \frac{\sqrt{K}}{\sqrt{K}+1}$$

$$\frac{1}{1-p} = \sqrt{K} + 1$$

部分排水系统，平衡时

$$\frac{dp}{dt} = k_1 c_0[(1-p)^2 - \frac{pn_w}{c_0 K}] = 0$$

则

$$\bar{X_n} = \frac{1}{1-p} = \sqrt{\frac{c_0 K}{pn_w}} \propto \sqrt{\frac{K}{n_w}}$$

The polymer is proportional to the square root of the equilibrium constant and inversely proportional to the sq

- The small polyesterification reaction in Changshu requires high temperature and reduced pressure to ful
- Polyamide formation reaction, k=400 can allow higher moisture content under slightly reduced pressur
- When K is large, the requirement for the degree of polymerization is not high; for example, the prepoly

2.5.2 The Influence of the Number of Functional Groups on the Degree of Polymerization

The above discussion is based on the condition of equimolar amounts of the two groups. To meet this co

1) Monomer high-purity purification and precise quantification;

2) Two functional groups are present on the same monomer molecule, such as hydroxy acids and amino acids

3) Binary amines and dicarboxylic acids first form salts.

In many practical operations, often on the basis of these measures, a certain binary mono mer is slightly in excess, or a small amount of mono-functional substance is additionally added to block the end groups, to control the degree of polymerization. For this purpose, the followin g two parameters are introduced:

$$r = \frac{N_a}{N_b}$$

$$q = \frac{(N_b - N_a)/2}{N_a/2} = \frac{1-r}{r}$$

$$r = \frac{1}{q+1}$$

Among them, N_a and N_b are the initial numbers of functional groups a and b in the system, respectively.

1. 在 2-2 体系中基团数不相等，设反应体系为

$$n\,aAa + m\,bBb \longrightarrow bB[AB]_n b$$

If the reaction extent of group a is p, then:

- a 的反应数为 $N_a p$，残留数为

$$N_a(1-p)$$

- b The number of reactions is also $N_a p$, and the number of residues is

$$N_b - N_a p$$

- The total number of residual end groups in the system is

$$N = N_a + N_b - 2N_a p$$

Since each macromolecular chain contains two end groups, the total number of macromolecules is half th

$$\frac{N_a + N_b - 2N_a p}{2}$$

The total number of structural units in the system is

$$\frac{N_a + N_b}{2}$$

Therefore, the degree of aggregation equals the number of structural units divided by the total number of

$$\bar{X}_n = \frac{(N_a + N_b)/2}{(N_a + N_b - 2N_a p)/2} = \frac{1+r}{1+r-2rp}$$

Consider two extreme cases:

1) 当 $r =$ 为 1 或 $q =$ 为 0 时,

$$\bar{X}_n = \frac{1}{1-p}$$

2) 当 $p =$ 为 1 时,

$$\bar{X}_n = \frac{1+r}{1-r}$$

When $r =$ is 1 and $p =$ is 1, theoretically the degree of polymerization tends to infinity, f orming an infinitely large macromolecule; however, in reality, p cannot be strictly equal to 1.

2. 2-2 体系中基团数相等时加入单官能团物质 若原本为两基团数相等的 2-2 体系，另外加入 N_b' 个单官能团物质 C_b，则有

$$r = \frac{N_a}{N_b + 2N_b'}$$

$$q = \frac{2N_b'}{N_a}$$

In the formula, “ 2 ” means: the excess of b functional groups caused by a monofunctional substance C_b is equivalent to the excess caused by a molecule bBb containing two b functional groups. Because in the excess bBb, only one group b participates in the end-capping, while the other group does not play a role in further chain growth, it can therefore be regarded as similar to a monofunctional substance C_b.

At this time there is also

$$\bar{X}_n = \frac{1+r}{1+r-2rp} = \frac{q+2}{q+2(1-p)}$$

It can be seen that slight measurement imbalances or the addition of a small amount of monofunctional end-capping agent can significantly affect the degree of polymerization of the polycondensation reaction. Therefore, in a polycondensation system, the purity of the raw materials, the stoichiometric ratio of functional groups, and the amount of end-capping agent must be strictly controlled.

2.6 Molecular Weight Distribution of Linear Condensates

2.7 Body Shrinking and Gelation Effects

2.8 Implementation methods of condensation and step-growth polymerization

2.9 重要的缩聚物和其他逐步缩聚物